

In[*]:= **NLODES =**

$$\{D[x[t], t] == V0 + V1 \beta - V2 + V3 + k f y[t] - k x[t], D[y[t], t] == V2 - V3 - k f y[t],$$

$$D[z[t], t] == \beta V4 - V5 - \epsilon z[t]\} /. \left\{V2 \rightarrow VM2 \frac{x[t]^2}{k2^2 + x[t]^2},\right.$$

$$\left. V3 \rightarrow VM3 \frac{x[t]^m}{kx^m + x[t]^m} \frac{y[t]^2}{ky^2 + y[t]^2} \frac{z[t]^4}{kz^4 + z[t]^4}, V5 \rightarrow VM5 \frac{z[t]^p}{k5^p + z[t]^p} \frac{x[t]^n}{kd^n + x[t]^n} \right\}$$

Out[*]=

$$\left\{x'[t] == V0 + V1 \beta - k x[t] - \frac{VM2 x[t]^2}{k2^2 + x[t]^2} + k f y[t] + \frac{VM3 x[t]^m y[t]^2 z[t]^4}{(kx^m + x[t]^m) (ky^2 + y[t]^2) (kz^4 + z[t]^4)},\right.$$

$$y'[t] == \frac{VM2 x[t]^2}{k2^2 + x[t]^2} - k f y[t] - \frac{VM3 x[t]^m y[t]^2 z[t]^4}{(kx^m + x[t]^m) (ky^2 + y[t]^2) (kz^4 + z[t]^4)},$$

$$\left. z'[t] == V4 \beta - \epsilon z[t] - \frac{VM5 x[t]^n z[t]^p}{(kd^n + x[t]^n) (k5^p + z[t]^p)} \right\}$$

In[*]:= **tmax = 20;**

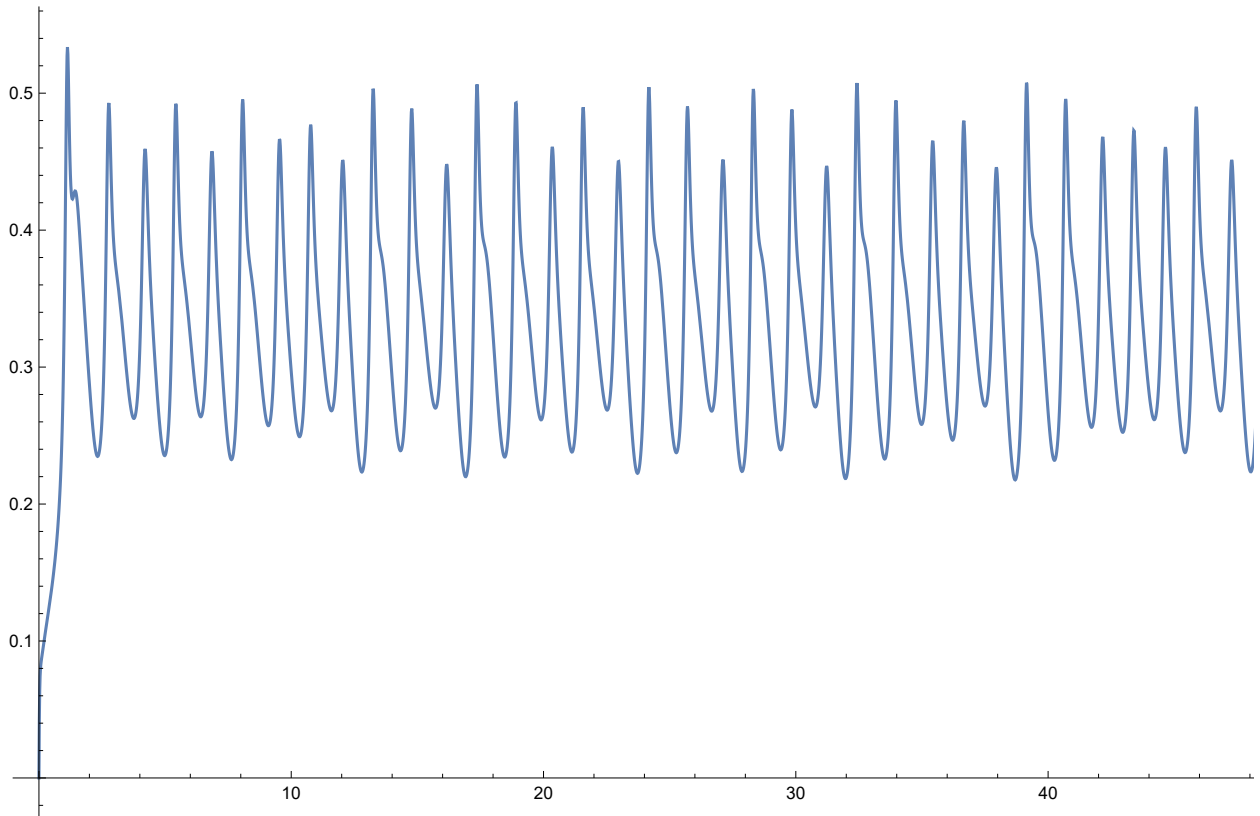
Chaos

```

In[ ]:= VvalsSS = {V0 → 2, V1 → 2, V4 → 3, kf → 1, k → 10, β → 0.65, ε → 13, VM2 → 6, k2 → 0.1,
  VM3 → 30, m → 2, kx → 0.6, ky → 0.3, kz → 0.1, VM5 → 50, k5 → 0.3194, kd → 1, p → 1, n → 4};
solSS1 = NDSolve[Join[NLODEs /. VvalsSS, {x[0] == 0, y[0] == 0, z[0] == 0}],
  {x[t], y[t], z[t]}, {t, 0, tmax}];
xv1 = x[t] /. solSS1;
Plot[xv1, {t, 0, tmax}, PlotRange → All]

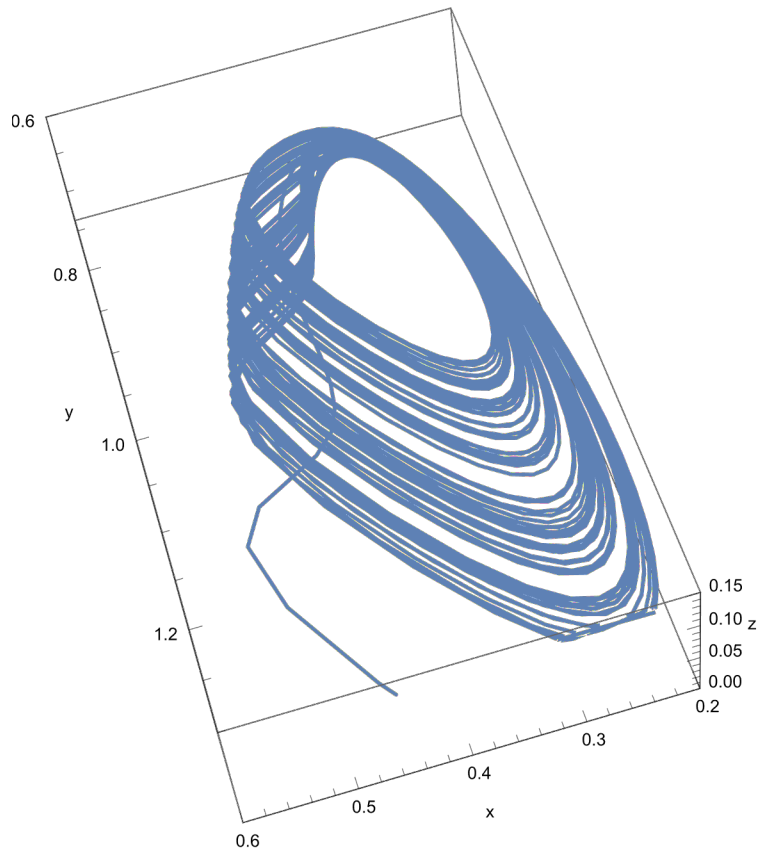
```

Out[]=



```
In[ ]:= ParametricPlot3D[{x[t] /. solSS1[[1]], y[t] /. solSS1[[1]], z[t] /. solSS1[[1]]}, {t, 0, tmax},
  AxesLabel -> {"x", "y", "z"}, PlotRange -> {{0.2, 0.6}, {0.6, 1.3}, {0, 0.15}}]
```

```
Out[ ]:=
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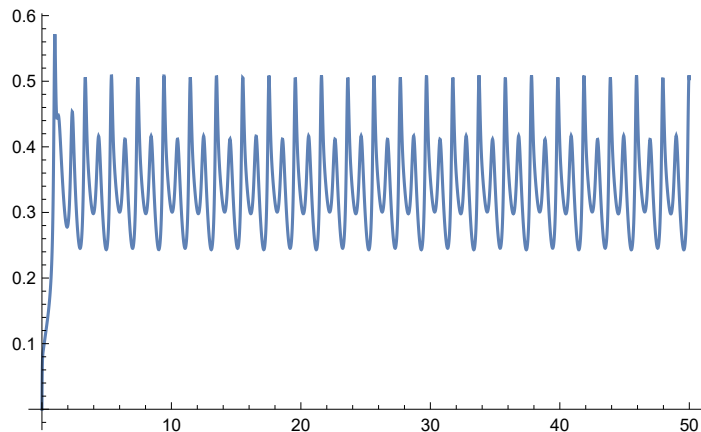
Period Doubling

```

In[ ]:= VvalsSS = {V0 → 2, V1 → 2, V4 → 3, kf → 1, k → 10, β → 0.7, ε → 13, VM2 → 6, k2 → 0.1, VM3 → 30,
  m → 2, kx → 0.6, ky → 0.3, kz → 0.1, VM5 → 50, k5 → 0.3194, kd → 1, p → 1, n → 4};
solSS1 = NDSolve[Join[NLODEs /. VvalsSS, {x[0] == 0, y[0] == 0, z[0] == 0}],
  {x[t], y[t], z[t]}, {t, 0, tmax}];
xv1 = x[t] /. solSS1;
Plot[xv1, {t, 0, tmax}, PlotRange → All]

```

Out[]:=

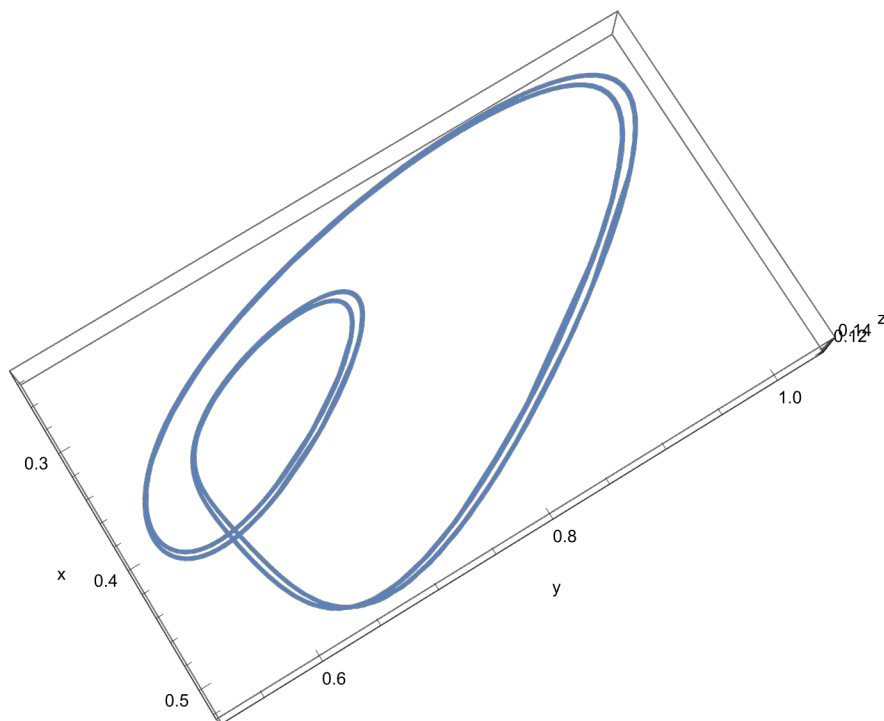


```

In[ ]:= ParametricPlot3D[{x[t] /. solSS1[[1]], y[t] /. solSS1[[1]], z[t] /. solSS1[[1]]}, {t, 10, tmax},
  AxesLabel → {"x", "y", "z"} (*, PlotRange → {{0.2, 0.6}, {0.6, 1.3}, {0, 0.15}} *)]

```

Out[]:=



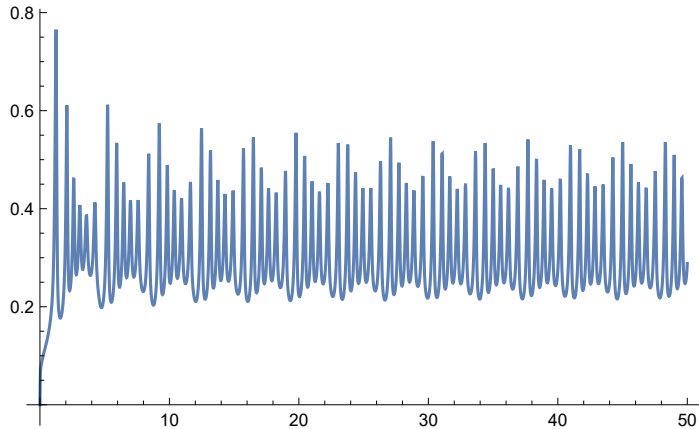
Quasi-Periodic

```

In[ ]:= VvalsSS = {V0 → 2, V1 → 2, V4 → 5, kf → 1, k → 10, β → 0.51, ε → 0.1, VM2 → 6, k2 → 0.1,
  VM3 → 20, m → 2, kx → 0.5, ky → 0.2, kz → 0.2, VM5 → 30, k5 → 0.3, kd → 0.5, p → 2, n → 4};
solSS1 = NDSolve[Join[NLODEs /. VvalsSS, {x[0] == 0, y[0] == 0, z[0] == 0}],
  {x[t], y[t], z[t]}, {t, 0, tmax}];
xv1 = x[t] /. solSS1;
Plot[xv1, {t, 0, tmax}, PlotRange → All]

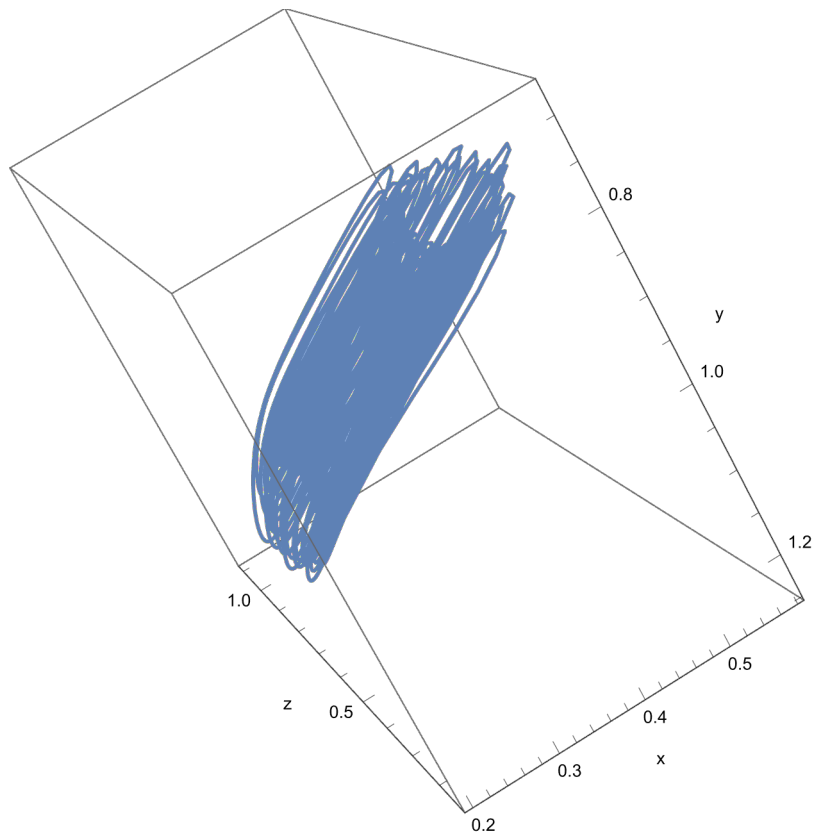
```

Out[]=



```
In[ ]:= ParametricPlot3D[{x[t] /. solSS1[[1]], y[t] /. solSS1[[1]], z[t] /. solSS1[[1]]}, {t, 10, tmax},  
  AxesLabel → {"x", "y", "z"} (*, PlotRange → {{0.2, 0.6}, {0.6, 1.3}, {0, 0.15}} *)]
```

Out[]=



Birhythmicity

```

In[ ]:= VvalsSS = {V0 → 2, V1 → 2, V4 → 3, kf → 1, k → 10, β → 0.52, ε → 11, VM2 → 6, k2 → 0.1,
  VM3 → 30, m → 2, kx → 0.6, ky → 0.3, kz → 0.1, VM5 → 50, k5 → 0.3194, kd → 1, p → 1, n → 4};
solSS1 = Table[NDSolve[
  Join[NLODEs /. VvalsSS, {x[0] == RandomReal[{0, 1000}], y[0] == RandomReal[{0, 100}],
    z[0] == RandomReal[{0, 100}]}], {x[t], y[t], z[t]}, {t, 0, tmax}], {i, 1, 10}];
xv1 = Table[x[t] /. solSS1[[nn]], {nn, 1, 10}];
Plot[xv1, {t, 0, tmax}, PlotRange → All]

```

Out[]:=



```

In[ ]:= t11 = Table[{x[t] /. solSS1[[n1]], y[t] /. solSS1[[n1]], z[t] /. solSS1[[n1]]}, {n1, 1, 10}];

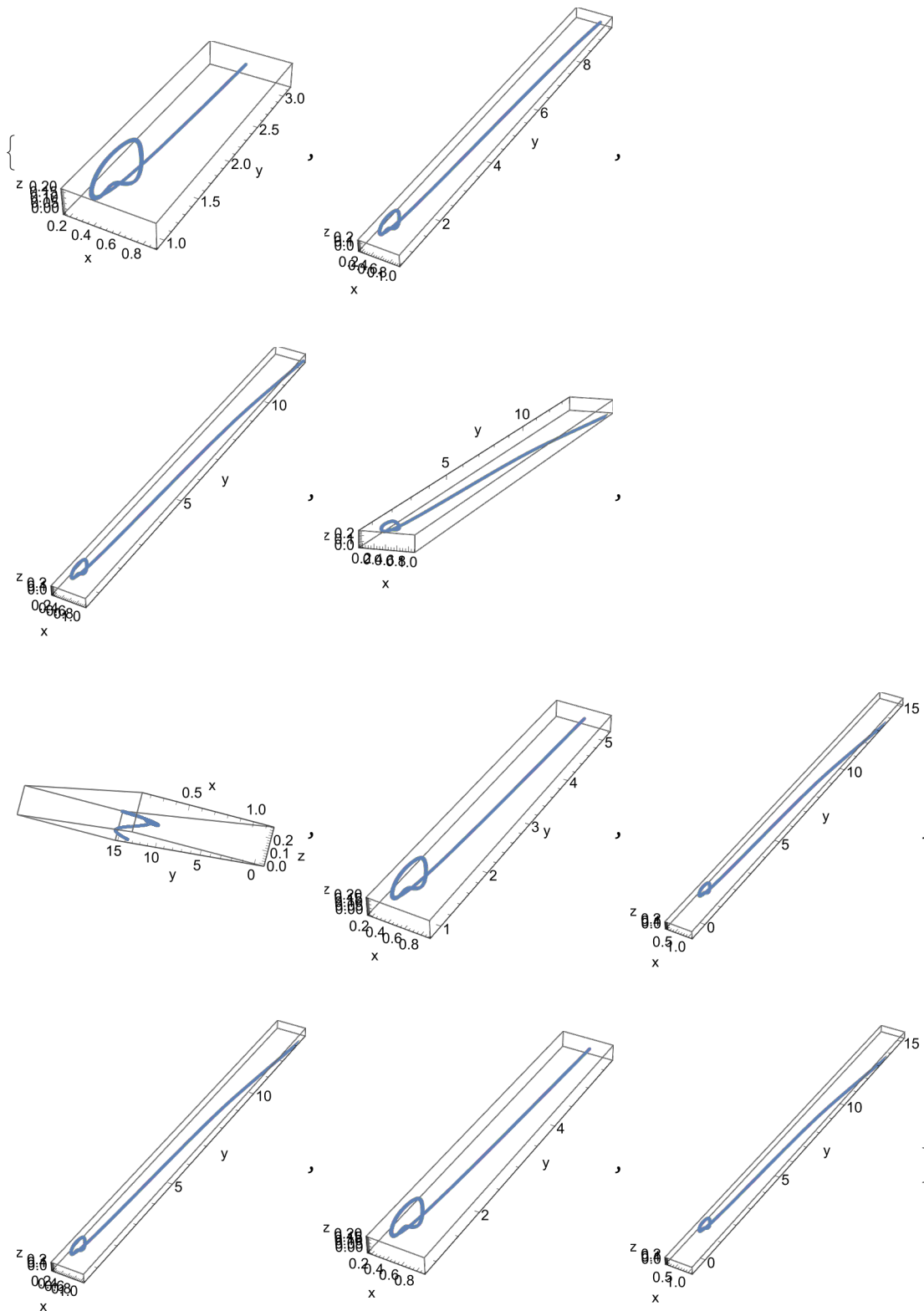
```

```

In[ ]:= Table[ParametricPlot3D[t11[[m1]], {t, 0, tmax}, AxesLabel -> {"x", "y", "z"}
  (*, PlotRange -> {{0.2, 0.6}, {0.6, 1.3}, {0, 0.15}} *)], {m1, 1, 10}]

```

Out[]=

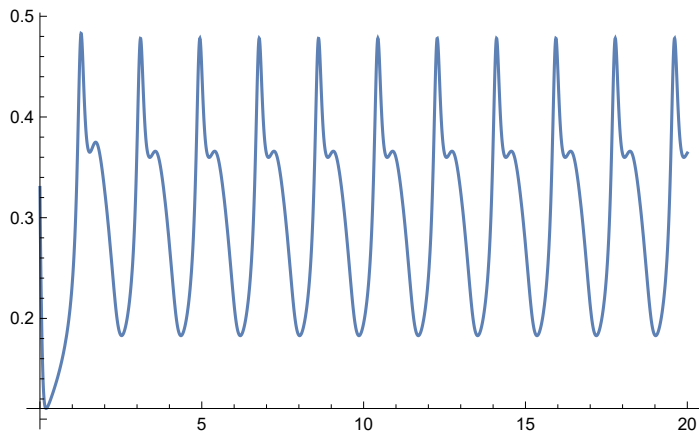


```

In[ ]:= VvalsSS = {V0 → 2, V1 → 2, V4 → 3, kf → 1, k → 10, β → 0.52, ε → 11, VM2 → 6, k2 → 0.1,
  VM3 → 30, m → 2, kx → 0.6, ky → 0.3, kz → 0.1, VM5 → 50, k5 → 0.3194, kd → 1, p → 1, n → 4};
solSS1 = NDSolve[Join[NLODEs /. VvalsSS, {x[0] == 0.330, y[0] == 0.2, z[0] == 0.2}],
  {x[t], y[t], z[t]}, {t, 0, tmax}];
xv1 = x[t] /. solSS1;
Plot[xv1, {t, 0, tmax}, PlotRange → All]

```

Out[]:=



```

In[ ]:= ParametricPlot3D[{x[t] /. solSS1[[1]], y[t] /. solSS1[[1]], z[t] /. solSS1[[1]]}, {t, 10, tmax},
  AxesLabel → {"x", "y", "z"} (*, PlotRange → {{0.2, 0.6}, {0.6, 1.3}, {0, 0.15}} *)]

```

Out[]:=

